

Enterprise Predictive Analytics Engine

Modern Data Stack Architecture Plan & Portfolio Blueprint (SQL + DuckDB + Python + Streamlit)

1. Executive Summary & Objective

This blueprint defines the enterprise-grade **Modern Data Stack (MDS)** architecture for the **Enterprise Predictive Analytics Engine**. Built on 113,425 e-commerce orders from the Olist Brazilian marketplace, this design elevates the project into a comprehensive **Analytics Engineering** and **Decision-Support Platform** combining standard SQL data modeling (DuckDB), advanced statistical modeling (scikit-learn, Prophet), and interactive BI dashboarding (Streamlit).

2. Modern Data Stack Architecture Overview

Architecture Layer	Technology / Tool	Core Responsibility & Value
1. Raw Storage	9 Relational CSV Tables	Transactional data spanning orders, items, payments, reviews, customers, sellers, and geolocation.
2. In-Process OLAP & SQL	DuckDB + SQL Layer	Executes relational joins, multi-item order deduplication (ROW_NUMBER), RFM CTE modeling, and SLA delay metrics.
3. Machine Learning	scikit-learn & Prophet	Supervised dissatisfaction risk classification (Random Forest), K-Means RFM clustering (k=4), and 6-month revenue forecasting.
4. Caching & Gateway	Parquet + Streamlit Cache	Columnar Parquet stores and in-memory caches (@st.cache_data / @st.cache_resource) for sub-second page loads.
5. Business Intelligence	Streamlit (8 Pages)	Executive KPIs, interactive real-time risk simulator, cohort retention heatmaps, and SQL data explorer.

3. Detailed SQL Analytics Models (sql/ Directory)

The SQL layer consists of 5 modular models showcasing advanced SQL proficiency: Common Table Expressions (CTEs), Window Functions (ROW_NUMBER, LAG), timestamp arithmetic (DATEDIFF), and conditional aggregation.

- **Model 1: 01_schema_and_views.sql:** Base transactional view performing multi-table LEFT JOINs across orders, items, payments, customers, and translated categories with NULL coalescing.
- **Model 2: 02_order_deduplication.sql:** Uses ROW_NUMBER() OVER (PARTITION BY order_id) to eliminate multi-item order duplicate payments and review scores, ensuring 100% accurate order-level revenue calculations.
- **Model 3: 03_rfm_customer_segmentation.sql:** Multi-CTE query calculating Recency (days to snapshot), Frequency (COUNT DISTINCT orders), and Monetary spend (SUM payments) grouped by customer_unique_id.
- **Model 4: 04_delivery_performance.sql:** Calculates logistics SLA deviation (DATEDIFF delivered vs. estimated), total transit duration, and generates the binary low_review target.
- **Model 5: 05_monthly_kpis_and_mom.sql:** Calculates monthly gross revenue, order volume, AOV, and utilizes LAG() OVER (ORDER BY month) to compute Month-over-Month (MoM) revenue growth rates.

```
-- Sample: 03_rfm_customer_segmentation.sql (CTE & Window Logic) WITH customer_orders AS ( SELECT
customer_unique_id, order_id, purchase_timestamp, total_payment_value FROM v_orders_deduplicated ) SELECT
customer_unique_id, DATEDIFF('day', MAX(purchase_timestamp), TIMESTAMP '2018-09-01 00:00:00') AS recency,
COUNT(DISTINCT order_id) AS frequency, ROUND(SUM(total_payment_value), 2) AS monetary FROM customer_orders GROUP BY
customer_unique_id;
```

4. Machine Learning & Predictive Engines

Model / Task	Algorithm & Config	Key Benchmark / Findings
Customer Segmentation	K-Means (k=4) + StandardScaler on RFM features.	96.8% of customers buy only once (55.5% Recent, 41.3% Lapsed). Loyal repeat buyers (3.1%) spend 2.1x more (R\$ 428 vs R\$ 198).
Dissatisfaction Risk Prediction	Random Forest Classifier (balanced, depth=10).	83.5% Accuracy, 0.430 F1 Score , 43.4% Precision. Delivery delays and transit duration drive 76.7% of all bad reviews .
Multi-Grain Revenue Forecast	Prophet Additive Time Series (90% CI, 6 periods).	Marketplace revenue projected to grow +19.4% over 6 months (R\$ 1.66M to R\$ 1.98M/mo). Granular forecasts for top 5 categories & states.

5. Business Value & Commercial Strategy

- **Shift Focus from Acquisition to Retention:** Converting the 55.5% Recent One-Time Buyers into repeat buyers via automated 30–60 day replenishment triggers offers the highest ROI growth lever.
- **Pre-Review Operational Interventions:** Real-time dissatisfaction scoring flags delayed shipments before delivery, allowing customer support to issue proactive tracking updates and vouchers.
- **Granular Financial Allocation:** Multi-grain forecasting enables regional logistics teams in SP, RJ, and MG to optimize warehouse capacity and inventory stocking ahead of seasonal demand.

6. How to Pitch This Architecture in Interviews

The 30-Second Interview Pitch:

'I built an Enterprise Predictive Analytics Engine using a Modern Data Stack architecture on 113,000 marketplace transactions. I used SQL and DuckDB to model raw relational data, eliminating revenue double-counting and engineering RFM features. I built an 83.5% accurate Random Forest model that flags at-risk deliveries before bad reviews occur, and projected 6-month revenues using Prophet. I deployed the full solution in an 8-page Streamlit decision dashboard backed by automated Pytest suites.'

7. Rapid-Fire Technical Cheat Sheet

Key Metric / Feature	Value / Finding
Total Dataset Size	113,425 order items 96,478 unique orders 93,358 customers
Customer Churn Dynamic	96.8% single-purchase rate 3.1% repeat buyers (spend 2.1x more)
Champion Classifier	Random Forest: 83.5% Accuracy 43.4% Precision 42.6% Recall 0.430 F1
Primary Dissatisfaction Driver	Delivery Delay vs. SLA (46.2%) + Transit Time (30.5%) = 76.7% Power
6-Month Revenue Projection	+19.4% Growth (R\$ 1.66M to R\$ 1.98M/month with 90% Confidence Bounds)
SQL & Analytics Stack	DuckDB, 5 Modular SQL scripts (CTEs, Window Functions), Parquet, Pytest